Chao Zhang

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EDUCATION

University of Chicago (Chicago, IL) Aug. 2017 – Jun. 2022

- PhD in Molecular Engineering | GPA 3.86/4.0
- Coursework: Modelling & Numerical Methods, Statistical Mechanics, Machine Learning, Algorithms, Python Programming, AB Testing

Zhejiang University (Hangzhou, China) Aug. 2013 -- Jun. 2017

- BS in Physics (Chu Kochen Honors College) | GPA 3.7/4.0 (top 5)

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL, Java

ML/AI Frameworks: PyTorch, TensorFlow, Transformer, Large Language Models (LLM), RAG

Tools & Platforms: Linux, AWS, Google Cloud, Spark, MongoDB, Django, MySQL, Git, Mercurial

EXPERIENCE

Meta Platforms | Research Scientist Jun. 2022 – Present

Remote

- Fine-tuned Llama3 model with PetaBytes of text/image data to adapt it to WhatsApp-specific harm detection tasks, and built account investigating AI agents with functionalities of dashboard search/review and database queries, achieved 30% improvement with verifiable evidence
- Developed Gemini-based RAG system to incorporate the latest violation patterns, enhancing traditional ML-based scam detection recall by 20% while maintaining same precision
- Engineered machine learning models from scratch using multi-modal inputs (text, image, numerical) to detect malicious accounts at scale, protecting billions of platform users
- Built end-to-end ML infrastructure for harm detection, managing the entire lifecycle from data collection and feature processing to model training and deployment
- Analyzed large-scale traffic data to evaluate model performance, delivering actionable insights that improved ML models and reduced platform harm

Seagate | Machine Learning Engineer Intern Nov. 2020 – Oct 2021

Remote

- Built multi-task classification deep learning architectures from scratch based on Auto-encoder and Transformer and had the models deployed in the factory with 30% efficiency improvement
- Applied various sampling and weighting methods in deep learning models to deal with high dimension and highly imbalanced industrial data sets
- Managed and oversaw the work of 5 other PhD students with CNN, GNN, ensemble model, anomaly detection, and data augmentation projects
- Drove external research collaborations on joint AI research for manufacturing applications with top universities including UMN and UCLA

University of Chicago | PhD researcher Aug. 2017 – Jun 2022

Pritzker school of Molecular Engineering

- Researched and developed a state-of-the-art, adaptive AI diagnosis system using a combination of numerical, image, and language models (BERT) on terabytes of medical records.
- Engineered and deployed the full-stack system, including a Django/MySQL web application on Google Cloud and a user-friendly Android app (Java/MongoDB) used in two hospitals in Pakistan.
- Led a multi-disciplinary team of medical scientists and software engineers, coordinating international collaborations to build and deploy the AI system.
- Presented research and applications at multiple academic conferences and won a Top 50 award in the Microsoft Azure AI Hackathon for the system's application.

AWARDS & LEADERSHIP

Award: Chinese Physics Olympiad (CPHO), Province-level First Prize (Top 0.1%).
Certificates: (coursera) Deep learning specialization, RAG, Agentic AI, DataCamp Data Scientist Track
Leadership: Led a team at the University of Chicago Chinese Students and Scholars Association, organizing seminars, forums, and other activities for over 2,000 people (2017-2019).

SELECTED FIRST AUTHOR PUBLICATIONS

- **Chao Zhang** "Advanced Text Mining Techniques and the Applications in Clinics, Chemistry and Manufacturing" The University of Chicago, 2022
- **Chao Zhang**, Jaswanth Yella, Yu Huang, Sthitie Bom "Learning from Failures in Large Scale Soft Sensing" IEEE International Conference on Big Data, 2021
- **Chao Zhang,** Jaswanth Yella, Yu Huang, Xiaoye Qian, Sergei Petrov, Andrey Rzhetsky, Sthitie Bom "Soft sensing transformer: hundreds of sensors are worth a single word" IEEE International Conference on Big Data, 2021
- **Chao Zhang**, Sthitie Bom "Auto-encoder based model for high-dimensional imbalanced industrial data" International Conference on Neural Information Processing, 2021
- **Chao Zhang**, Hanxin Zhang, Atif Khan, Ted Kim, Olasubomi Omoleye, Oluwamayomikun Abiona, Amy Lehman, Olufunmilayo I. Olopade, Pedro Lopes, Andrey Rzhetsky "Lightweight Mobile Diagnostic System for Global Lower-Resource Areas" PLOS Global Public Health, submitted
- **Chao Zhang**, Ming Yang, Bao Yang, Kian Ping Loh, Jianlan Wu and Yuan Ping Feng "Abnormal behavior of potassium adsorbed phosphorene" International Journal of Computational Materials Science and Engineering, 2017